DEVANSH TYAGI

Aspiring Data Scientist

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SUMMARY

I am an aspiring Data Scientist and a final-year Computer Science student specialized in AI/ML. I have engineered predictive neural network models to extract actionable insights and deployed real-time ML solutions using FastAPI and Next.js. My skills in statistical analysis and data visualization have enabled me to solve complex real-world problems effectively.

SKILLS

ML & AI Tools: Scikit-learn | TensorFlow.js | Keras | CNNs | OpenCV | Neural Networks | Model Deployment (FastAPI)
Data Analytics & Maths: Exploratory Data Analysis (EDA) | Microsoft Excel (XLOOKUP, VLOOKUP, Pivot Tables) | Statistical Modeling | Multivariate Calculus | Linear Algebra | Probability & Statistics | Optimization
Programming: Python (Pandas, NumPy, Matplotlib) | JavaScript | C++ | PHP
Web Development & DBs: Next.js | Node.js | React | MERN Stack | REST APIs | MySQL | MongoDB | Mongoose

EDUCATION

Bachelor of Science (Hons.) Computer Science — Expected Aug 2026
Ramanujan College, University of Delhi — *Delhi, India*

- Scored an aggregate of **9.27 CGPA**.
- Relevant Coursework: Data Structures, Algorithms, DBMS, Multivariate Calculus, Linear Algebra.

Senior Secondary & High School Education — 2020 – 2022
Divya Public School, Shamli — *Uttar Pradesh, India*

- Awarded School Topper for both 10th (94.4%) and 12th (93%) standard board examinations.
- Completed specialized coursework in Physics, Chemistry, Mathematics, and Computer Science.

PROJECTS

Materials Informatics & Machine Learning | **crystalogix.netlify.app**

- Developed and authored a research paper, currently under review in npj Computational Materials, proposing a hybrid XGBoost–ensemble architecture for accurate prediction of electronic bandgaps in inorganic crystals.
- Engineered a custom, GPU-accelerated pipeline using Materials Project data and a hybrid classification–regression workflow to efficiently screen promising materials candidates.
- Built and deployed CrystaLogiX, an end-to-end AI application using FastAPI, Redis, and Next.js, enabling users to interact with the trained model through real-time bandgap predictions.

Delhi-NCR Air Quality Analysis & Visualization

- Led a data science project performing **Exploratory Data Analysis** on 10 years of NASA Giovanni satellite data.
- Improved data consistency by 70% using advanced preprocessing on large-scale environmental datasets.
- Identified a 15% decrease in CO and 20% increase in NO_x using **time-series forecasting**.

PopcornPick – Movie Recommendation Platform | **popcornpickapp.netlify.app**

- Built a custom ML recommendation model, improving content relevance and personalization by 30%.
- Deployed RESTful APIs using FastAPI to serve real-time recommendations with sub-second response times.

Interactive Single-Layer Perceptron | **perceptron-playground.vercel.app**

- Visualized real-time weight updates and convergence to enhance understanding of linear classifiers.
- Achieved **100% classification accuracy** on linearly separable datasets.
- Designed a user-friendly interactive dashboard with sliders and input fields, reducing model training and tuning time by 20%.

MNISTic – Handwritten Digit Predictor | **mnistic.netlify.app**

- Trained a deep learning model on the MNIST dataset, achieving 98%+ validation accuracy.
- Enabled real-time predictions from canvas input, making the classification process 70% faster.

WORK EXPERIENCE

Full Stack Developer [MERN] — Feb 2025 – Apr 2025
LOUDER, Lajpat Nagar — *New Delhi, India*

- Rebuilt the legacy ticketing system using the **MERN stack**, improving scalability.
- Reduced average page load time by 35–45% by optimizing API calls and frontend rendering.
- Enhanced database query efficiency, decreasing response latency by 30% through schema redesign and indexing.